TPC-H queries — reconstructed 2026-10-12

These 21 `.sql` files are **in-repo on purpose.** `collect_query.py` and `extract_plan_features.py` previously pointed at `/Users/harshithj/Main/Archive/OtherFiles/patched_queries`, outside version control. That directory no longer exists, so the exact query text behind the entire dataset was lost. These files are a reconstruction, validated against the artifacts the originals left behind.

Never move these out of the repo again. The query text is as much a part of the dataset as the timings.

Validation method

The local `tpch` database is byte-exact standard SF1 on every table (lineitem 6,001,215; orders 1,500,000; customer 150,000; part 200,000; partsupp 800,000; supplier 10,000; nation 25; region 5), and `results/plan_structure_features.csv` was generated from it with the *original* queries. That file therefore fingerprints each original query: output cardinality, node count, plan depth, total cost and summed actual rows.

Each reconstruction was checked against all five. Results: `results/query_reconstruction_validation.csv`.

Check	Result
Plan shape (`node_count` + `plan_depth`) exact	**21 / 21**
Output cardinality (`root_actual_rows`) exact	**19 / 21**
`rows_processed` within 1%	15 / 21 (within 5%: 20 / 21)
Verdict	**19 CONFIRMED, 2 APPROXIMATE**

Residual drift in `rows_processed` and `root_cost` is the ANALYZE/vacuum-history effect already documented in `results/buffer_bug_correction_report.md` — cost estimates and per-node actual row counts move with statistics and with how many parallel workers the executor happens to get. Output cardinality does not, which is why it is the primary fingerprint.

Two parameters recovered by search, not assumption

- **q11 uses `KENYA`, not the textbook `GERMANY`.** At the spec fraction 0.0001, GERMANY yields 1048 rows; the original yielded 1217. Sweeping all 25 nations, KENYA gives exactly 1217 — and then `rows_processed` (125,013), `node_count` (15), `plan_depth` (6) and `root_cost` (10,534.80) all match to the digit. Unambiguous.
- **q18 uses `having sum(l_quantity) > 314`,** not 300. 300 yields 57 rows; the original yielded
- Thresholds 313–316 all yield 9 and are indistinguishable (the constant sits in a HAVING, so the plan is identical); 314 was chosen as the midpoint of the spec's 312–315 range. All five fingerprint fields match exactly.

Two queries that remain APPROXIMATE — know this before using them

query	original	reconstruction	gap
q16	18,141 rows	18,314 rows	**+0.95%**
q20	193 rows	186 rows	**−3.63%**

Both have the correct structure and an exactly matching plan shape, and q20's total cost matches to 0.00%. Only a substitution constant differs.

- **q16**: the excluded brand and/or type prefix differs. An analytic sweep of all 25 brands × 30 spec type-prefixes against the standard size set `(49,14,23,45,19,3,36,9)` produced **no** pair yielding 18,141 — so the eight-element size set itself must differ, and C(50,8) is too large to search.
- **q20**: the colour and/or nation differs. 15 (nation, colour) pairs from a 32-word colour list yield exactly 193 rows and all give the identical plan and cost, but none reproduces the original's `rows_processed` of 5,426,542 (closest: INDIA/peru at +0.42%, IRAQ/navy at −0.56%). TPC-H builds `p_name` from 92 colour words; the true one is outside the list tried.

Consequence: any *new* measurement on q16 or q20 — elasticity, plan features, laptop-as-dev timings — describes a query very slightly more selective or less selective than the one in `tpch_dataset_corrected_v2.csv`. Plan shape, join structure and data volume are preserved, so resource-elasticity conclusions carry over; absolute runtimes for these two do not. Every derived artifact must carry the flag.

Reproducing the validation

```
python3 scripts/collection/validate_queries.py
```